WaveDSNet: A Wavelet-Enhanced Discrepancy Semantic Network for Water Change Detection in SAR Imagery

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***Abstract*—Water change detection in synthetic aperture radar (SAR) imagery is important for inundation extent delineation and river course monitoring. However, existing methods still face three core challenges: inherent speckle noise interference, insufficient extraction of semantic-structural features, and the lack of dedicated benchmarks for evaluating high-resolution structural details. To address these issues, a wavelet-enhanced discrepancy semantic network (WaveDSNet) and the Xidian University SAR Water Change Detection (XDU-SWCD) dataset are proposed for water change detection in SAR imagery. WaveDSNet consists of two key components: first, a wavelet-based noise suppression (WNS)-embedded backbone performs wavelet-subband-aware denoising to suppress speckle while preserving structural cues; second, a semantic-boundary collaborative refinement (SBCR) module aggregates bi-temporal features into difference and semantic streams, and further coordinates them through bidirectional interaction and adaptive gating, with auxiliary boundary-aware supervision for fine-grained changed-water boundary localization. Furthermore, the XDU-SWCD dataset is constructed as a focused high-resolution benchmark for SAR water change detection. It combines fine-grained water-change structures with representative confusing nonwater regions, enabling a rigorous evaluation of boundary delineation and robustness to low-backscatter pseudo-changes. The experiments on XDU-SWCD show that WaveDSNet achieves stronger overall performance than nine representative change detection methods, with an overall IoU of 0.7796 and an overall $F1$-score of 0.8590. Zero-shot cross-dataset evaluation on public benchmarks further demonstrates its promising transferability. These results suggest the effectiveness of both the proposed framework and the XDU-SWCD benchmark for SAR water change detection. Our code and XDU-SWCD dataset will be publicly available at https://github.com/gosling123456/WaveDSNet.git**

***Index Terms*—Discrepancy semantic network, hard-negative scenarios, high-resolution dataset, water change detection, wavelet dynamic convolution.**

I. Introduction

WATER change detection, which characterizes hydrological dynamics using remote sensing imagery, is fundamental to inundation extent delineation [1], [2] and river course monitoring [3], [4]. While optical imagery offers rich spectral information and intuitive interpretability, it is inherently constrained by atmospheric conditions under dense cloud cover and low illumination. These conditions frequently coincide with rapid hydrological events, rendering optical imagery insufficient for reliable monitoring in emergency scenarios. In contrast, synthetic aperture radar (SAR) is largely unaffected by cloud cover and illumination conditions, providing all-weather, day-and-night imaging capabilities [5]. Consequently, SAR serves as an important data source for tracking dynamic hydrological processes under extreme meteorological conditions.

Despite these advantages, the coherent imaging mechanism of SAR introduces unique challenges for water change detection. These challenges motivate extensive research on robust feature extraction methods. Traditional water change detection methods in SAR imagery typically rely on handcrafted features [6], [7] and a multistage pipeline involving filtering, segmentation, and postprocessing [8]. These methods are not end-to-end and remain sensitive to imaging conditions [9], often requiring extensive expert intervention for parameter tuning [10]. Furthermore, the reliance on isolated pixel-wise representations inherently limits the integration of broad contextual information. Consequently, these methods often yield fragmented detection results that fail to preserve the spatially coherent patterns of water expansion or recession [8], [11].

Recent deep learning models, including convolutional neural network (CNN)-based [12], [13] and transformer-based [14], [15], [16] methods, have significantly advanced remote sensing change detection. Meanwhile, recent studies have explored lightweight and efficient architectures for SAR flood mapping and change detection [17], [18], [19]. These studies highlight the importance of balancing detection performance and computational efficiency for practical flood-response applications. However, SAR water change detection remains challenging due to the coherent imaging characteristics of SAR and the

Received 27 April 2026; revised 16 August 2026 and 8 September 2026; accepted 17 September 2026. Date of publication 22 September 2026; date of current version 30 September 2026. This work was supported in part by the "Scientists + Engineers" Project of Qin Chuangyuan in Shaanxi Province of China under Grant 2024QCY-KXJ-152; in part by the Key Industry Chain Technology Research and Development General Project of Xi'an under Grant 2024JH-CLYB-0047; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2024CY2-GJHX-17; in part by the National Natural Science Foundation of China under Grant 62371373, Grant 62271377, Grant 62401418, and Grant 62501450; and in part by the Natural Science Basic Research Program of Shaanxi under Grant 2026JC-YBQN-0792. *(Corresponding authors: Zhongle Ren; Kai Wang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3736079

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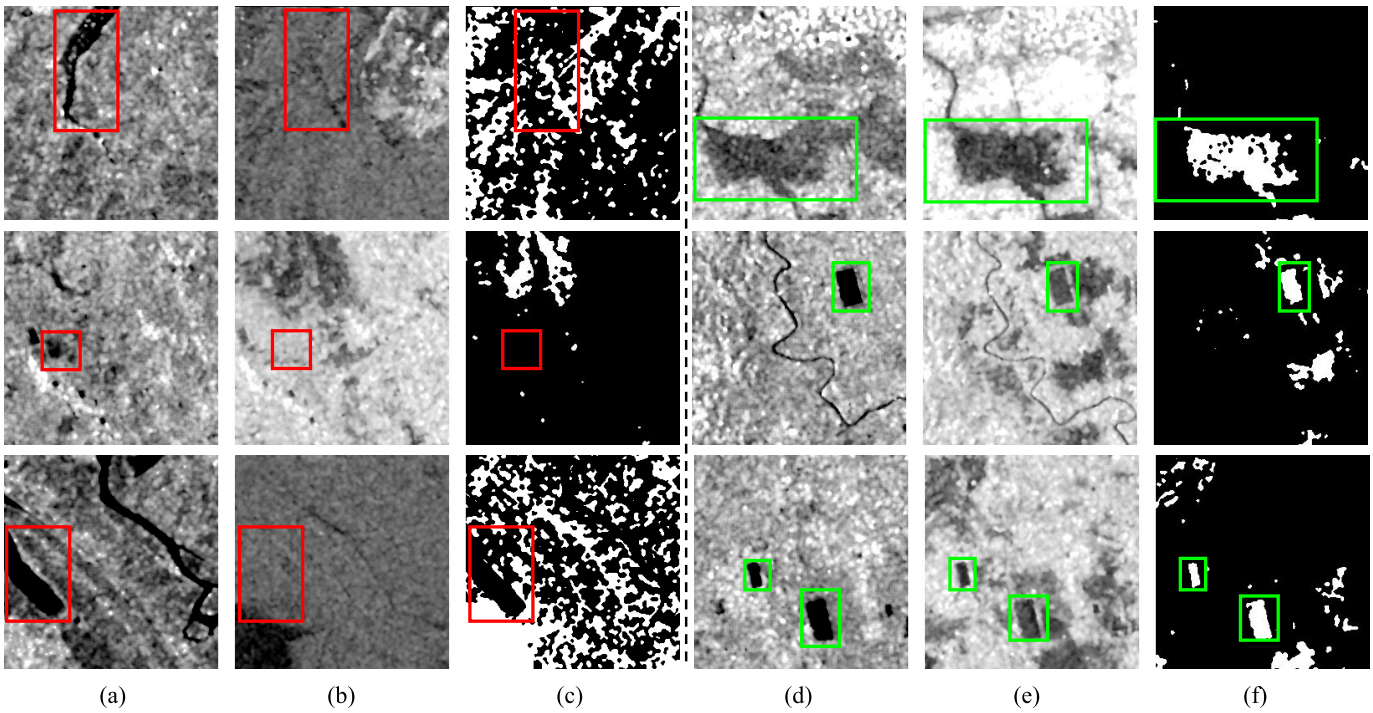


Fig. 1. Illustration of typical annotation errors caused by semiautomated labeling pipelines on S1GFloods. (a) and (d) Postchange images. (b) and (e) Prechange images. (c) and (f) Annotation masks. The left (highlighted in red) demonstrates omission errors, where fine-grained water or weak boundaries are missed by the algorithm. The right (highlighted in green) displays commission errors, where environmental interference is incorrectly labeled as water change.

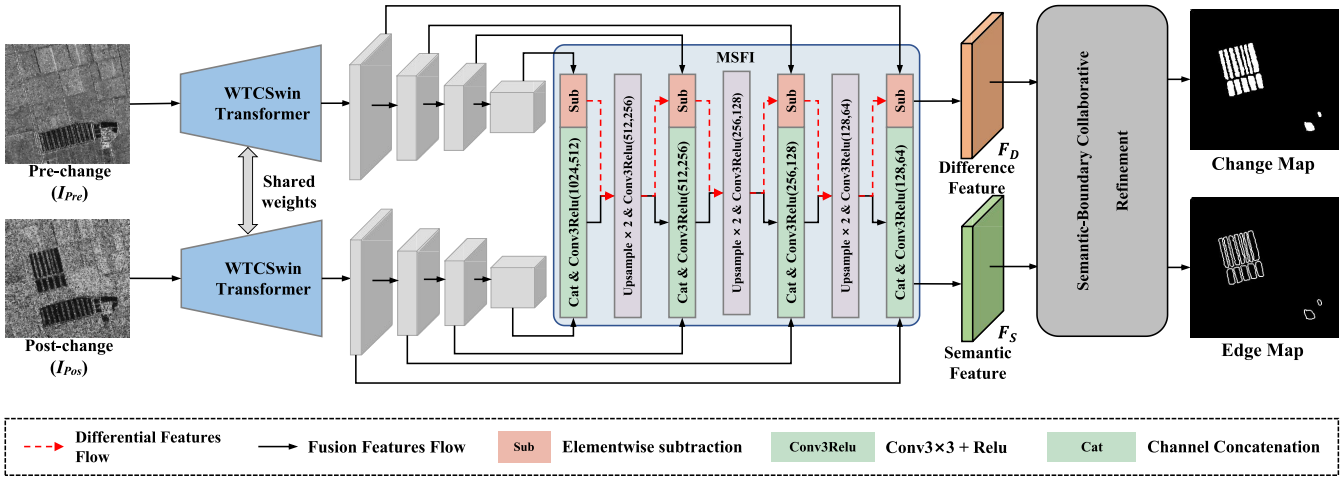


Fig. 2. Overall architecture of the proposed WaveDSNet.

limitations of current benchmarks. First, multiplicative speckle noise severely interferes with feature extraction, especially in low-texture homogeneous regions and boundary transition zones, making it difficult to distinguish stochastic speckle artifacts from genuine water-change boundaries. Although spatial-domain CNNs and transformers can suppress irregular responses through feature aggregation, excessive smoothing may attenuate weak water changes, narrow tributaries, and fragmented shorelines, whereas directly preserving high-frequency responses may amplify speckle noise. Frequency-domain methods can provide useful spectral priors at different frequency scales, but global frequency representations offer limited spatial localization. Wavelet decomposition provides localized multiscale representations by organizing feature responses into spatially aligned subbands; however, speckle artifacts and genuine boundaries may still coexist within the same subband. Moreover, low-backscatter targets such as shadows, airport runways, irrigation zones, and wetlands may produce abnormal responses similar to those of changed water, making it difficult to suppress noise and confusing responses while preserving meaningful structural details. Second, under the side-looking imaging geometry and complex scattering characteristics of SAR, existing methods still struggle to sufficiently coordinate temporal discrepancy features

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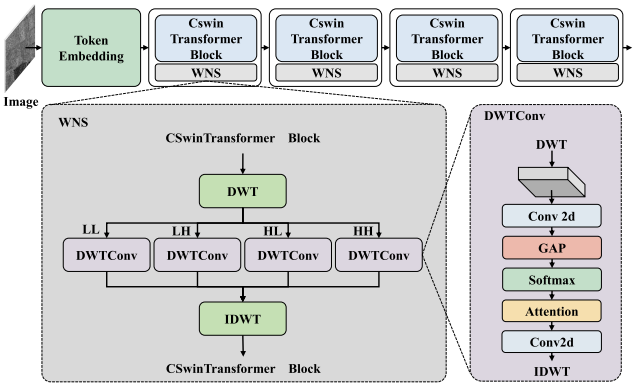


Fig. 3. Architecture of the WTCSwinTransformer.

TABLE I
PUBLIC DATASET INFORMATION

Dataset	Sensor	Res.	Category	KHNS*	Pairs
Bern [22]	ERS-2	30m	Low	-	4
Chaohu-1 [23]	Sentinel-1	10m	Medium	-	4
Ottawa [22]	Radarsat-1	10m	Medium	-	4
San Francisco [22]	ERS-2	20m	Medium	-	1
Sardinia [22]	Landsat-5	30m	Low	-	4
Sulzberger-1 [22]	ENVISAT	10m	Medium	-	1
Sulzberger-2 [22]	ENVISAT	10m	Medium	-	1
Yellow-River1 [22]	Radarsat-2	4m	High	-	4
Yellow-River2 [22]	Radarsat-2	4m	High	-	4
Yellow-River3 [22]	Radarsat-2	4m	High	-	4
Yellow-River4 [22]	Radarsat-2	4m	High	-	4
S1GFloods [24]	Sentinel-1	10m	Medium	4	5360
VarFloods-G [25]	Sentinel-1	10m	Medium	-	2186
VarFloods-P [25]	Sentinel-1	10m	Medium	-	2186
XDU-SWCD (Ours)	**GF-3**	**3m**	**High**	**6**	**3,203**

* **Category**: High (<5m), Medium (5-20m), Low (>20m).

* **KHNS**: Kinds of hard-negative samples.

and semantic-structural information, which often leads to pseudo-changes and fragmented water-change boundaries [20], [21]. Temporal discrepancy features can highlight potential changed regions but may be disturbed by unstable scattering responses, while semantic features may lack sufficient detail for fine-grained boundary localization. Consequently, effectively exploiting their complementarity while maintaining boundary continuity remains challenging. Third, most existing public datasets rely on medium-resolution sensors (Table I, see Section IV) [22], [23], [24], [25], which limits the capture of fine-grained object details and hinders rigorous evaluation of boundary preservation. Moreover, existing benchmarks are not specifically designed to evaluate high-resolution boundary fidelity and curated low-backscatter hard-negative scenarios. Some datasets may also contain coarse or noisy annotations around weak boundaries and confusing regions. As illustrated by representative cases from S1GFloods in Fig. 1 (see Section IV), semiautomated labeling pipelines may miss fine boundary details or mistakenly assign change labels to low-backscatter regions. These limitations make it difficult to rigorously assess semantic robustness and boundary fidelity in complex SAR scenes.

To address these issues, a wavelet-enhanced discrepancy semantic network (WaveDSNet) and the XDU-SWCD dataset are proposed for water change detection in SAR imagery. WaveDSNet consists of two key components: a wavelet-based noise suppression (WNS)-embedded backbone and a semantic-boundary collaborative refinement (SBCR) module. The WNS-embedded backbone performs wavelet-subband decomposition followed by content-adaptive dynamic convolution, so that speckle-dominated responses and boundary-sensitive structures can be refined in a more targeted manner. The SBCR module takes the aggregated difference and semantic features as two complementary streams, establishes bidirectional interaction between them through cross-attention, and further applies adaptive gating to determine whether semantic detail or discrepancy evidence should dominate prediction at different locations. The refined representation is then regularized by auxiliary boundary-aware supervision to improve contour continuity and fine-grained changed-water localization. Furthermore, the XDU-SWCD dataset is constructed as a focused high-resolution benchmark for SAR water change detection. It contains 3203 valid bi-temporal GF-3 image pairs at 3-m resolution, collected from four large-scale scenes under diverse temporal change settings, including Dry→Wet, Wet→Wet, and Wet→Dry conditions. In addition, it explicitly incorporates six representative hard-negative scenarios, namely, roads, buildings, airport runways, irrigation zones, shadows, and wetlands, which frequently exhibit low-backscatter responses similar to water. These characteristics provide fine-grained water-change samples under diverse seasonal conditions together with representative confusing nonwater targets, thereby enabling more rigorous evaluation of boundary fidelity and semantic robustness in complex SAR scenes.

The main contributions of this work are summarized as follows.

1) A new framework, termed WaveDSNet, is proposed for SAR water change detection. By combining speckle-robust representation learning with semantic–difference collaboration and boundary-aware refinement, it improves the detection of changed water regions in semantically confusing and boundary-sensitive SAR scenes.
2) A WNS-embedded backbone is introduced to perform content-adaptive wavelet-subband refinement, which facilitates differentiated refinement of speckle-dominated responses and structure-sensitive subband components and enhances the representation of weak water changes, narrow tributaries, and fragmented shorelines in SAR imagery.
3) An SBCR module is proposed to coordinate discrepancy cues and semantic-structural information through bidirectional interaction and adaptive gating, while fur-

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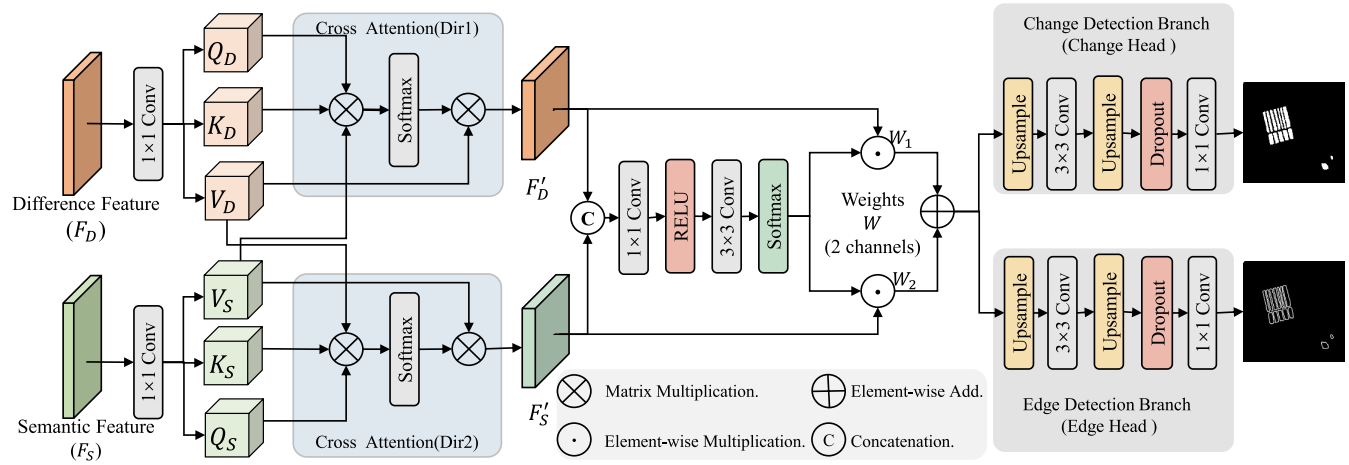


Fig. 4. Structure of the SBCR module.

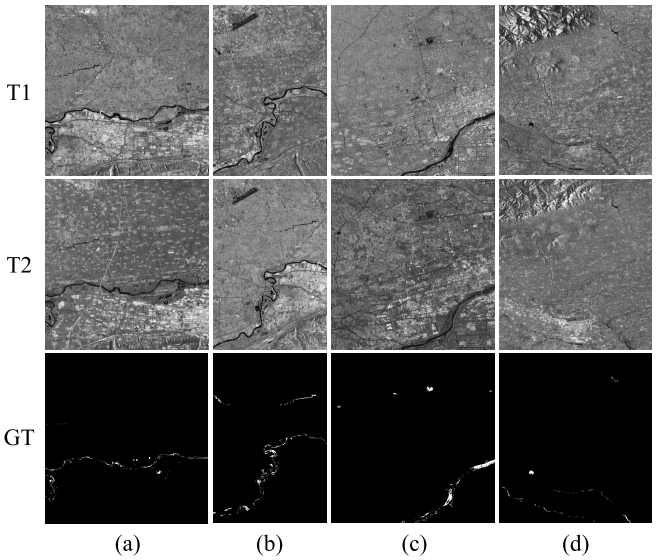


Fig. 5. Illustration of the XDU-SWCD dataset. For each geographic area, the first row (T1) shows the prechange images, the second row (T2) shows the postchange images, and the third row presents the GT. (a) Xi'an, (b) Xi'an and Xianyang, (c) Xianyang, and (d) Xianyang and Tongchuan.

ther imposing auxiliary boundary-aware supervision. This design refines feature representations in boundary regions and improves contour continuity and fine-grained water-change localization.

4) XDU-SWCD is constructed as a high-resolution SAR water change detection benchmark containing 3203 GF-3 image pairs and representative hard-negative scenarios. It provides a dedicated benchmark for evaluating high-resolution boundary fidelity and robustness to pseudo-changes in low-backscatter hard-negative scenes.

The remainder of this article is organized as follows. Section II reviews related SAR change detection methods. Section III presents the architecture and key components of WaveDSNet. Section IV introduces the construction, characteristics, and annotation strategy of the proposed dataset. Section V reports experimental results and analyses. Section VI discusses the limitations and practical implications of the proposed method. Finally, Section VII summarizes the findings and outlines future directions.

II. Related Work

A. Change Detection Methods in RS Imagery

Over the past decades, remote sensing change detection methodologies have evolved from rudimentary algebraic operations to shallow machine learning techniques, and more recently, to deep learning paradigms characterized by hierarchical feature representations. Traditional approaches, such as image differencing, ratioing, change vector analysis (CVA) [26], and principal component analysis (PCA)-based transformations [27], are computationally efficient. However, they inherently lack spatial contextual modeling, making them sensitive to radiometric variations and registration errors. To improve discrimination, machine learning-based methods introduce handcrafted features combined with classifiers like support vector machine (SVM) [28] and multilayer perceptron (MLP) [29], as well as fuzzy c-means (FCMs) clustering [30]. Nevertheless, their performance remains constrained by the limited representation capability of engineered features.

With the advent of deep learning, change detection has shifted toward end-to-end representation learning. CNN-based Siamese models [12], [13] effectively capture local spatial cues. However, standard CNNs primarily emphasize local context and may be less effective in modeling long-range dependencies. To address this, transformer-based architectures [31], [32] and hybrid CNN–transformer models have gained prominence by leveraging global self-attention mechanisms. Recent advancements also focus on utilizing bi-temporal information more effectively, employing frequency-domain priors [33], [34] or joint fusion-difference learning [35] to preserve high-frequency boundaries and suppress noise.

While generic deep learning frameworks have shown success, applying them to water change detection in SAR imagery presents unique challenges due to speckle noise, temporal inconsistency, and complex boundary structures. Early SAR-specific approaches, including thresholding [6], graph cuts

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Fig. 6. Illustration of challenging hard-negative samples that resemble water in SAR imagery. To verify robustness against semantic confusion, typical interference targets are explicitly marked: roads, buildings, airport runways, irrigation zones, shadows, and wetlands.

[36], and level sets [37], offer interpretable solutions but degrade sharply under strong speckle and spatial heterogeneity. Weakly supervised pipelines [38] attempt to mitigate annotation scarcity but inevitably inherit noise from the pseudo-label generation process.

Current state-of-the-art methods in water change detection in SAR imagery largely adapt the aforementioned deep learning paradigms to address these domain-specific issues. Attention mechanisms and specialized modules have been integrated into CNN structures [39] to enhance representation quality. Furthermore, transformer-based designs like TransUNet++SAR [40] and DAM-Net [24] exploit global dependencies to suppress noise-driven false positives. Despite the above progress, existing methods still have difficulty in jointly handling speckle suppression, boundary preservation, and pseudo-change reduction in high-resolution SAR water change detection. Traditional filtering methods may reduce speckle but often blur narrow water structures, while CNN-based and transformer-based models mainly operate in the spatial domain, where speckle responses and meaningful boundaries can remain entangled. Although frequency-domain and wavelet-related strategies provide useful priors, the joint modeling of subband-aware denoising and semantic-level pseudo-change suppression remains insufficient. This motivates the proposed WNS-embedded backbone and SBCR module, which perform localized spatial-frequency refinement while coordinating discrepancy cues and semantic-structural information.

B. Datasets for Water Change Detection in SAR Imagery

Early benchmarks like Bern and Ottawa [22], [23] have laid the foundation for water change detection in SAR imagery, but their limited sample sizes are insufficient for training and evaluating modern deep models. To alleviate data scarcity, larger-scale benchmarks such as S1GFloods [24] and VarFloods [25] are introduced, establishing useful baselines for flood mapping. However, these Sentinel-1-based datasets still face two key limitations for fine-grained water change detection.

1) *Performance Ceiling Caused by Scene Simplicity:* Existing benchmarks often contain relatively easy samples with clear contrast between changed-water regions and the surrounding unchanged background. Several recent methods have achieved $F1$-scores exceeding 90% [24], [25] on specific large-scale flood benchmarks, particularly in relatively high-contrast scenarios. This ceiling effect reduces the discriminative power of the benchmark for comparing advanced methods.
2) *Resolution Constraints and Semantic Ambiguity:* Restricted by the 10-m resolution of Sentinel-1 [24], [25], current datasets suffer from mixed-pixel effects that blur fine topological structures such as narrow tributaries. Furthermore, the lack of textural detail exacerbates semantic ambiguity, making low-backscatter nonwater targets (e.g., radar shadows and irrigation zones) difficult to distinguish from water [41].

Motivated by these limitations, XDU-SWCD is introduced in this work as a focused high-resolution benchmark for more rigorous evaluation of water change detection in SAR imagery.

III. Methodology

A. Overall Structure of WaveDSNet

Fig. 2 shows the overall framework of WaveDSNet, which consists of two key components: a WNS-embedded backbone and an SBCR module. First, the WNS-embedded backbone decomposes features into localized wavelet subbands and applies content-adaptive refinement to suppress speckle-dominated responses while preserving structure-sensitive cues. Then, the bi-temporal multiscale features are aggregated and reformulated into a difference stream and a semantic stream by the improved multiscale feature interaction (MSFI) [16]. Finally, the SBCR module facilitates interaction between the two streams, adaptively fuses them, and introduces boundary-aware supervision to improve pseudo-change suppression and boundary localization.

B. WNS-Embedded WTCSwinTransformer Backbone

In SAR water change detection, early feature refinement should address both frequency entanglement and spatial heterogeneity. Wavelet decomposition is adopted because, compared with global transforms such as the *discrete cosine transform* and the *discrete Fourier transform*, it is more suitable for localized spatial-frequency analysis, which is important for jointly handling speckle suppression and boundary preservation in remote sensing imagery [42]. By decomposing features into one low-frequency component and three directional high-frequency subbands, the wavelet transform separates coarse regional content from boundary-sensitive structures, making subsequent refinement more targeted [43], [44]. However, subband decomposition alone is still insufficient, because the relative distributions of speckle responses and structural information vary across input features and wavelet subbands. For this reason, dynamic convolution is further introduced. Unlike standard convolution with shared kernels, dynamic convolution generates input-dependent filters for each wavelet subband, enabling the filtering behavior to

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adapt to the content distribution of the current feature map [45], [46]. Therefore, wavelet decomposition is used to provide localized frequency separation, while dynamic convolution is used to provide content-adaptive subband refinement.

As illustrated in Fig. 3, the WTCSwinTransformer organizes feature extraction into hierarchical stages. Designed as a Siamese architecture with shared weights, it processes the bi-temporal input images $I_{Pre}, I_{Pos} \in \mathbb{R}^{B\times C\times H\times W}$ independently, where B denotes the batch size, C is the number of input channels, H represents the height of the input images, and W stands for the width of the input images. Each stage adopts a serial "extract-then-refine" architecture, comprising a CSWin transformer block [47] and a proposed WNS module. First, the CSWin block captures global semantic contexts via cross-shaped window self-attention [47]. Following the spatial modeling of the CSWin block, the WNS module is integrated to perform wavelet-domain refinement. The core motivation is to address the overlap between stochastic speckle responses and fine-grained structural details in the high-frequency domain. Standard spatial convolutions struggle to disentangle these conflicting signals. To facilitate the disentanglement of these signals, the module applies a single-level discrete wavelet transform (DWT), using zero-padding to handle boundary conditions, to decompose the input feature map $X \in \mathbb{R}^{B\times C\times H\times W}$ into four distinct subbands: one low-frequency component ($Y_{LL} \in \mathbb{R}^{B\times C\times H/2\times W/2}$) and three high-frequency components ($Y_{LH}, Y_{HL}, Y_{HH} \in \mathbb{R}^{B\times C\times H/2\times W/2}$)

$$(Y_{LL}, Y_{LH}, Y_{HL}, Y_{HH}) = \mathrm{DWT}(X). \tag{1}$$

Traditional CNNs use spatially invariant convolutional kernels, applying the same convolutional kernel across the entire image [48], [49]. However, SAR imagery is affected by multiplicative noise whose intensity varies with signal strength [50], [51]. To break this invariance, dynamic convolution is introduced, allowing the generated kernel W_s^{dyn} to adapt to the content of each input subband [48], [49]. Consequently, the WNS module provides frequency-specific and content-aware denoising behavior that is not explicitly modeled in standard transformers. For each subband feature Y_s, a channel-attention projection generates adaptive weights

$$A_s = \mathrm{Softmax}(\mathrm{GAP}(\phi(Y_s))) \tag{2}$$

where $\phi(\cdot)$ denotes a 1×1 convolutional projection, GAP signifies global average pooling, and $A_s \in \mathbb{R}^K$ denotes the kernel weighting vector. The dynamic convolution kernel W_s^{dyn} is synthesized through the weighted fusion of a set of $K = 4$ learnable base kernels $\{W_s^k\}_{k=1}^K$

$$W_s^{\mathrm{dyn}} = \sum_{k=1}^{K} A_s^k W_s^k. \tag{3}$$

This design allows the filtering behavior of different subbands to adapt to their content characteristics, facilitating speckle suppression and structural preservation. Each subband undergoes feature transformation via its corresponding dynamic kernel

$$\hat{Y}_s = \mathrm{Conv}\left(Y_s, W_s^{\mathrm{dyn}}\right). \tag{4}$$

Through this input-dependent modulation, the filtering process is adapted to the frequency characteristics of the current subband feature. Specifically, the kernel mixture is conditioned on the global content statistics of each subband. This enables the WNS module to adjust its filtering behavior according to the relative prominence of irregular speckle responses and boundary-sensitive high-frequency structures.

The processed subbands are subsequently reconstructed into spatial features through inverse wavelet transform (IDWT)

$$\hat{X} = \mathrm{IDWT}\left(\hat{Y}_{LL}, \hat{Y}_{LH}, \hat{Y}_{HL}, \hat{Y}_{HH}\right). \tag{5}$$

This wavelet-domain processing pipeline ensures structural preservation while enhancing discriminative features in change-sensitive regions.

Let $\{\mathbf{X}_l^{\mathrm{pre}}\}_{l=1}^L$ and $\{\mathbf{X}_l^{\mathrm{pos}}\}_{l=1}^L$ denote the multiscale features extracted from I_{Pre} and I_{Pos}, respectively, where l indexes the feature level. After WNS-based refinement, two complementary feature streams are constructed for subsequent cross-temporal interaction.

First, the *difference feature* at the lth level is obtained by element-wise subtraction between the bi-temporal features

$$\mathbf{D}_l = \mathbf{X}_l^{\mathrm{pos}} - \mathbf{X}_l^{\mathrm{pre}} \tag{6}$$

which emphasizes cross-temporal discrepancy cues and highlights potentially changed regions.

Meanwhile, the *semantic feature* is constructed by concatenating the bi-temporal features and projecting them through a convolution block

$$\mathbf{S}_l = \phi_l\left(\mathrm{Cat}\left(\mathbf{X}_l^{\mathrm{pre}}, \mathbf{X}_l^{\mathrm{pos}}\right)\right) \tag{7}$$

where $\mathrm{Cat}(\cdot)$ denotes channel-wise concatenation, and $\phi_l(\cdot)$ denotes a 3×3 convolution followed by a ReLU activation. Compared with $\mathbf{D}_l$, $\mathbf{S}_l$ preserves richer object-level contextual semantics and boundary-sensitive structural information.

To obtain compact representations for the subsequent SBCR module, the multilevel difference and semantic features are further aggregated by hierarchical fusion

$$\widetilde{\mathbf{D}}_l = \mathcal{U}_l(\mathbf{D}_l), \quad \widetilde{\mathbf{S}}_l = \mathcal{U}_l(\mathbf{S}_l) \tag{8}$$

$$\mathbf{F}_D = \psi_D\left(\mathrm{Cat}\left(\widetilde{\mathbf{D}}_1, \widetilde{\mathbf{D}}_2, \ldots, \widetilde{\mathbf{D}}_L\right)\right) \tag{9}$$

$$\mathbf{F}_S = \psi_S\left(\mathrm{Cat}\left(\widetilde{\mathbf{S}}_1, \widetilde{\mathbf{S}}_2, \ldots, \widetilde{\mathbf{S}}_L\right)\right) \tag{10}$$

where $\mathcal{U}_l(\cdot)$ denotes upsampling to a common spatial resolution, and $\psi_D(\cdot)$ and $\psi_S(\cdot)$ denote convolutional fusion operators. In this way, $\mathbf{F}_D$ mainly captures cross-temporal discrepancy cues, whereas $\mathbf{F}_S$ retains region-level semantic context and fine-grained structural details.

C. SBCR Module

Although the WNS-embedded backbone alleviates speckle interference and preserves structural details, subsequent decoding still faces a tradeoff between change sensitivity and boundary fidelity. Existing change detection frameworks often rely either on a shared semantic/fused representation, e.g., transformer-based bi-temporal interaction schemes such as BIT and ChangeFormer, or on refined difference features that explicitly emphasize changed regions [31], [52], [53].

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The former is beneficial for contextual reasoning but may oversmooth fine boundaries, whereas the latter provides clearer positional responses but is more vulnerable to pseudo-changes in confusing regions [53], [54]. Therefore, instead of decoding from only one dominant representation, the semantic features and the difference features are further interactively fused and adaptively reweighted, so that the model can selectively rely on semantic detail or discrepancy evidence at different locations before subsequent boundary-aware supervision. The above unified refinement pipeline is shown in Fig. 4.

Given input features F_D, $F_S \in \mathbb{R}^{B\times C\times H\times W}$, the module first generates query, key, and value projections through convolutional layers

$$\begin{cases} Q_D = \mathrm{Conv}_{1\times1}(F_D) \\ K_S = \mathrm{Conv}_{1\times1}(F_S) \\ V_S = \mathrm{Conv}_{1\times1}(F_S) \end{cases} \quad \begin{cases} Q_S = \mathrm{Conv}_{1\times1}(F_S) \\ K_D = \mathrm{Conv}_{1\times1}(F_D) \\ V_D = \mathrm{Conv}_{1\times1}(F_D) \end{cases} \tag{11}$$

where Q, K, and V denote the query, key, and value projections, respectively. The subscripts D and S indicate that the corresponding projections are generated from the difference feature F_D and the semantic feature F_S, respectively. Specifically, Q_D, K_D, and V_D are projected from F_D, whereas Q_S, K_S, and V_S are projected from F_S.

The module computes attention in both directions to capture comprehensive feature relationships

$$F'_D = \mathrm{Softmax}\left(\frac{Q_D K_S^\top}{\sqrt{d_k}}\right) V_S \tag{12}$$

$$F'_S = \mathrm{Softmax}\left(\frac{Q_S K_D^\top}{\sqrt{d_k}}\right) V_D \tag{13}$$

where d_k denotes the dimension of key vectors, and the $\sqrt{d_k}$ term serves as a scaling factor to stabilize attention scores.

The outputs from both attention directions are then adaptively fused to obtain a refined representation

$$F_{\mathrm{fused}} = W_1 \odot F'_D + W_2 \odot F'_S \tag{14}$$

where the spatially adaptive weights W_1 and W_2 are generated from the concatenated features

$$[W_1, W_2] = \mathrm{Softmax}\left(\phi_g\left(\mathrm{Cat}\left(F'_D, F'_S\right)\right)\right). \tag{15}$$

Here, $\odot$ denotes element-wise multiplication, $\mathrm{Cat}(\cdot)$ denotes channel-wise concatenation, and $\phi_g(\cdot)$ denotes the gating function. Based on F_{fused}, two prediction heads with the same architecture but unshared parameters are introduced

$$\hat{Y}_{\mathrm{change}} = \mathcal{H}(F_{\mathrm{fused}}; \theta_c), \quad \hat{Y}_{\mathrm{edge}} = \mathcal{H}(F_{\mathrm{fused}}; \theta_e) \tag{16}$$

where $\mathcal{H}(\cdot;\theta)$ denotes a lightweight prediction head composed of convolution, dropout, and upsampling operations, while θ_c and θ_e denote the parameters for the change and edge branches, respectively. Although the two heads share the same architecture, their parameters are independent and they are optimized with different supervision targets. The change head predicts dense changed-water regions, whereas the edge head localizes sparse boundaries of changed-water regions. This design avoids forcing a single mapping to fit both region-level and boundary-level objectives, while the consistent head structure ensures that their functional distinction mainly arises from supervision rather than architectural differences. Since both heads take F_{fused} as input, the edge loss provides boundary-focused gradients to regularize the shared representation used by the change branch. During inference, only $\hat{Y}_{\mathrm{change}}$ is used as the final prediction, whereas $\hat{Y}_{\mathrm{edge}}$ is employed only during training as an auxiliary supervision signal.

D. Loss Function

To ensure robust gradient propagation and comprehensive supervision across both pixel-level and region-level constraints, a composite hybrid loss function is employed. This strategy combines the strengths of pixel-wise probabilistic supervision and global structural alignment. The binary cross-entropy (BCE) loss is utilized to optimize pixel-wise classification accuracy, ensuring stable convergence by penalizing probabilistic deviations at individual locations

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right]. \tag{17}$$

Complementarily, the Dice loss is incorporated to quantify the global similarity between the predicted change map and the ground truth (GT). This encourages the model to focus on structural coherence and region compactness rather than treating pixels in isolation

$$\mathcal{L}_{\mathrm{Dice}} = 1 - \frac{2|X\cap Y|}{|X|+|Y|} \tag{18}$$

$$= 1 - \frac{2\sum_i p_i y_i}{\sum_i p_i + \sum_i y_i + \epsilon} \tag{19}$$

where ϵ is a small positive constant set to 10^{-6} in all experiments to avoid division by zero and improve numerical stability. The combined loss integrates these complementary objectives

$$\mathcal{L}_{\mathrm{C}} = \mathcal{L}_{\mathrm{BCE}} + \mathcal{L}_{\mathrm{Dice}}. \tag{20}$$

To further refine boundary localization, edge-aware supervision is incorporated during training

$$\mathcal{L}_{\mathrm{final}} = \mathcal{L}_{\mathrm{C}}\left(Y_{\mathrm{change}}, \hat{Y}_{\mathrm{change}}\right) + \lambda\mathcal{L}_{\mathrm{C}}\left(Y_{\mathrm{edge}}, \hat{Y}_{\mathrm{edge}}\right) \tag{21}$$

where λ balances the contributions of change and edge tasks. The edge supervision target Y_{edge} is derived from the GT change mask Y_{change} using the Canny edge detector followed by a 3×3 morphological dilation. Since positive edge pixels are naturally sparse, $\hat{Y}_{\mathrm{edge}}$ is treated as an auxiliary supervision signal rather than a dominant optimization target. In our implementation, no additional postprocessing or output fusion is applied to the edge branch; instead, its influence is controlled by λ, while the combination of BCE and Dice losses helps alleviate the imbalance between sparse boundary pixels and nonedge regions. In this way, $\hat{Y}_{\mathrm{edge}}$ mainly provides boundary-specific regularization for water-change contour localization without destabilizing the optimization of the primary change detection branch.

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TABLE II
XDU-SWCD DATASET INFORMATION

Area	Time (T1 / T2)	Seasonal Transition	Sensor	Size	Res.	Pol.	Samples
Xi'an	2019-01 (Dry) / 2019-06 (Wet)	Dry → Wet	GF-3	7199 × 7516	3 m	HH	870
Xi'an & Xianyang	2019-06 (Wet) / 2020-09 (Wet)	Wet → Wet	GF-3	5490 × 8250	3 m	HH	726
Xianyang	2019-06 (Wet) / 2019-11 (Dry)	Wet → Dry	GF-3	6219 × 6473	3 m	HH	650
Xianyang & Tongchuan	2019-12 (Dry) / 2020-09 (Wet)	Dry → Wet	GF-3	7393 × 8266	3 m	HH	957

IV. XDU-SWCD DATASET

A. Dataset Positioning and Scope

The Xidian University SAR Water Change Detection (XDU-SWCD) dataset is designed as a *focused high-resolution benchmark* for water change detection in SAR imagery, rather than a universally comprehensive dataset covering all geographic regions, sensor types, and hydrological settings.

Its main objective is to provide a more rigorous evaluation protocol for fine-grained boundary preservation and robustness against confusing low-backscatter interference, rather than to compensate for methodological shortcomings. Accordingly, WaveDSNet addresses the modeling challenges of SAR water change detection, while XDU-SWCD provides a dedicated benchmark for evaluating performance in high-resolution and semantically challenging scenarios.

B. Dataset Composition and Statistics

XDU-SWCD contains 3203 valid bi-temporal image pairs acquired by the GF-3 satellite in HH polarization at a spatial resolution of 3 m. As shown in Fig. 5, all samples are cropped into nonoverlapping 256 × 256 patches from four large-scale scenes in Shaanxi Province, China, including the Weihe River basin and surrounding urban water systems in Xi'an, Xianyang, and Tongchuan. Table II summarizes the dimensions and temporal settings of the four source regions, which collectively cover Dry→Wet, Wet→Wet, and Wet→Dry conditions, representing water expansion, seasonal persistence, and water recession, respectively. Specifically, the Xi'an, Xi'an and Xianyang, Xianyang, and Xianyang and Tongchuan scenes contribute 870, 726, 650, and 957 samples, respectively. This design increases the diversity of hydrological change patterns under a unified sensor configuration.

The four original large-scale bi-temporal SAR scenes are first co-registered and annotated at the scene level. The dataset is divided into training, validation, and test subsets at an approximate ratio of 8:1:1 using a geographic-region-level splitting strategy. Specifically, each original scene is partitioned into spatially disjoint geographic regions, which are assigned to the three subsets before patch generation. Nonoverlapping 256 × 256 image pairs are subsequently cropped within the corresponding geographic partitions. This strategy avoids direct pixel overlap across subsets and reduces the risk of spatial context leakage caused by randomly mixing neighboring patches from the same local area. The resulting training, validation, and test sets contain 2,563, 320, and 320 image pairs, respectively. Since patch generation is performed after scene-level co-registration and annotation, the boundary labels are determined with reference to the complete large-scene context rather than being independently annotated within isolated patches. At the spatial resolution of XDU-SWCD, each patch covers approximately 768 × 768 m, providing sufficient local context for most water-change structures. Nevertheless, fixed-size cropping may still truncate large water bodies or transition boundaries near patch borders. Future work will investigate full-scene prediction and more patch-based processing strategies to better preserve contextual information.

C. Fine-Grained Scenarios and Hard-Negative Scenarios

Compared with medium-resolution public benchmarks such as S1GFloods [24], VarFloods [25], and other smaller datasets shown in Table I, XDU-SWCD is intended to evaluate fine-grained water change detection in SAR imagery under two specific challenges: the preservation of weak boundaries and semantic confusion in low-backscatter regions. First, the 3-m spatial resolution allows narrow tributaries, jagged shorelines, and fragmented boundaries of changed-water regions to remain visible, whereas such structures are often blurred by mixed-pixel effects in 10 m imagery. Consequently, the dataset is suitable for evaluating whether a model can localize thin changed structures rather than only coarse changed regions.

Second, Fig. 6 shows that XDU-SWCD includes six representative hard-negative scenarios involving roads, buildings, airport runways, irrigation zones, shadows, and wetlands. These regions are challenging because they may exhibit low-backscatter responses similar to water or geometric and scattering patterns resembling water changes in SAR imagery. Under the binary annotation protocol, pixels in these regions are assigned unchanged labels and participate in the standard pixel-wise supervision. Consequently, false activations over these regions are penalized by the standard pixel-wise loss. These hard-negative scenarios are not mutually exclusive semantic or image-level classes. XDU-SWCD is designed for binary water change detection, and a single bi-temporal image pair may contain changed-water structures together with one or more confusing nonwater regions. Therefore, increasing the coverage of fine-grained water-change structures does not necessarily result in a corresponding reduction in the number of image pairs containing hard-negative targets. It should be emphasized that the changed/unchanged class imbalance itself is not considered beneficial; it remains an inherent challenge because changed-water regions generally occupy a relatively small and spatially irregular portion of each image. The value

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of XDU-SWCD instead arises from its coverage of diverse water-change structures and representative confusing nonwater conditions under different temporal and geographic settings.

Temporally, the dataset spans from January 2019 to September 2020 and covers both hydrologically dry and wet periods. This temporal span introduces substantial intraclass variability. In particular, water bodies may expand during wet periods and shrink into fragmented pools during dry periods, while shallow-water and wet-soil transition zones often produce weak and ambiguous boundaries. As a result, XDU-SWCD is designed to test both region-level change recognition and boundary-level localization under realistic seasonal appearance variation.

D. Data Processing

Before patch generation, all bi-temporal SAR image pairs are aligned using their geographic coordinate information under a common spatial reference, and the final registration error is controlled to within 2 pixels to limit, although not completely eliminate, registration-induced boundary noise. This level of registration accuracy limits the impact of spatial offsets, which is particularly important because the benchmark focuses on fine-grained boundary evaluation, where even small offsets may introduce artificial boundary noise. To reduce the influence of extreme intensity outliers and normalize the dynamic range of SAR data, a statistical percentile-based linear stretching operation is applied

$$I_{8bit} = \text{clip}\left(\frac{I_{16bit} - P_1}{P_{99} - P_1} \times 255,\ 0,\ 255\right) \tag{22}$$

where P_1 and P_{99} denote the 1st and 99th percentiles of the pixel intensity distribution, respectively. In this work, these two percentiles are computed on a per-scene basis. This preprocessing suppresses extreme outliers caused by specular reflections or abnormal bright responses while preserving the relative radiometric structure needed for subsequent learning.

E. Annotation Strategy and Label Quality Control

In this work, a *water change* is defined as a spatial transition in water extent between two SAR acquisitions. The annotation focuses on changes in rivers, lakes, reservoirs, ponds, and other stable open-water bodies. Regions with transient radiometric darkening but without clear open-water extent transition are not labeled as change. For ambiguous cases such as shallow-water transition zones, wet soil, or seasonally flooded agricultural parcels, labels are assigned only when multisource evidence supports a clear transition of the water boundary; otherwise, they are marked as unchanged. High-quality pixel-level annotation is essential for this benchmark because the main purpose of XDU-SWCD is to evaluate the preservation of fine-grained changed-water boundaries. In this work, annotation quality is controlled through a multistage expert verification strategy rather than relying solely on semi-automated thresholding. First, each scene is independently annotated by two domain experts according to hydrological morphology, shoreline continuity, and bi-temporal consistency. Second, ambiguous regions are cross-validated using auxiliary optical imagery from the same or temporally adjacent period, together with historical inundation records and map references when available. Third, disagreements are reviewed by a senior researcher, who determines the final label after checking both spatial coherence and temporal plausibility. Here, two representative annotation noise patterns are considered. *Omission errors* refer to missed fine-grained changed regions, typically occurring around narrow tributaries, weak boundaries, and shallow-water transition zones under coarse preliminary labeling. *Commission errors* refer to falsely activated change labels over unchanged low-backscatter nonwater regions, such as shadows, dark roads, or irrigation-related surfaces.

V. Experiments

A. Experimental Settings

1) Training Parameters: The model is trained for 100 epochs with a batch size of 16. The maximum learning rate, weight decay, and task-balancing coefficient λ are set to 1×10^{-3}, 1×10^{-4}, and 0.7, respectively. Training is performed on one NVIDIA RTX 5090 GPU with 32 GB of memory, and num_workers is set to 16. AdamW [55] is adopted as the optimizer, and CosOneCycle [56] is used for learning-rate scheduling.

2) Metrics: To comprehensively evaluate the experimental results, this work adopts several commonly used evaluation metrics, including precision, IoU, $F1$-score, recall, and Cohen's Kappa. These metrics are reported for the changed class and for the overall change detection performance. Specifically, the subscript "1" denotes the metrics computed for the changed class, whereas metrics without this subscript denote the overall change detection performance.

3) Comparative Methods: In this work, nine representative change detection models are evaluated, encompassing both classical and recently proposed architectures. Classic CNN-based methods, including FC-EF [12], FC-Siam-Conc [12], FC-Siam-Diff [12], and DTCDSCN [13], serve as conventional baselines. Mainstream transformer-based models, namely, BIT [31] and ChangeFormer [52], reflect the progression toward attention-based mechanisms. In addition, several recent state-of-the-art models are included: EATDer [14], SFEARNet [15], and DDRL [16], which integrate advanced design strategies such as edge-aware refinement and domain adaptation.

B. Comparative Results

1) Results on XDU-SWCD: Table III and Fig. 7 present the quantitative and qualitative comparisons on XDU-SWCD. WaveDSNet achieves the best IoU_1, $F1\text{-score}_1$, Recall_1, overall IoU, overall $F1$-score, overall recall, and Kappa, while maintaining a competitive Precision_1 of 0.7775. Although DTCDSCN yields slightly higher Precision_1 and overall precision, its much lower Recall_1 indicates an overly conservative tendency that misses many changed regions. Compared with EATDer, the strongest competing method in terms of IoU_1 and $F1\text{-score}_1$, WaveDSNet still achieves clear improvements of 6.47 and 5.53 percentage points in IoU_1 and $F1\text{-score}_1$, respectively. In the representative visual examples, CNN-based models often suffer from false activations in confusing

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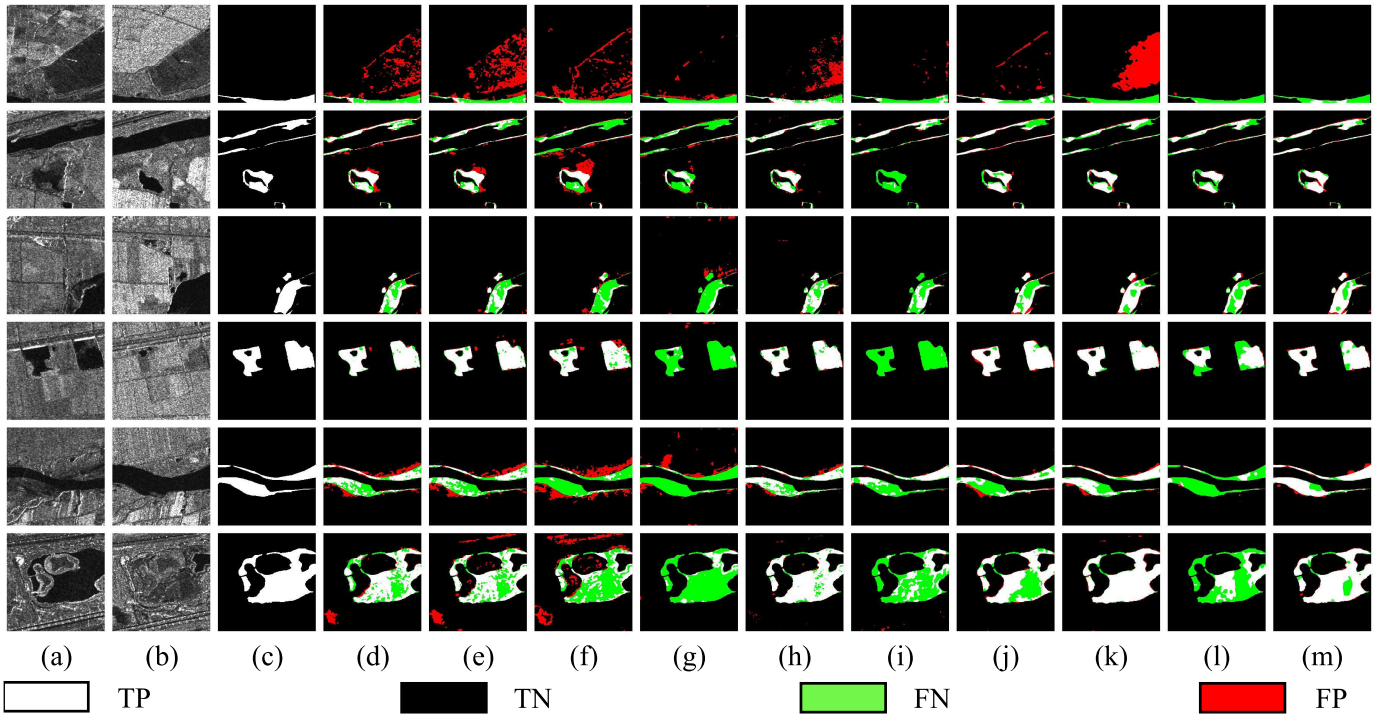


Fig. 7. Qualitative comparisons on the XDU-SWCD. (a) T1 image. (b) T2 image. (c) GT. (d) FC-EF. (e) FC-Siam-Conc. (f) FC-Siam-Diff. (g) BIT. (h) ChangeFormer. (i) DTCDSCN. (j) EATDer. (k) SFEARNet. (l) DDRL. (m) WaveDSNet.

TABLE III
PERFORMANCE COMPARISON OF CHANGE DETECTION METHODS ON XDU-SWCD

Method	Change Class Metrics				Overall Metrics				
	Precision_1	IoU_1	F1-score_1	Recall_1	Precision	IoU	F1-score	Recall	Kappa
FC-EF	0.5008	0.3926	0.5638	0.6451	0.7496	0.6941	0.7808	0.8211	0.5616
FC-Siam-Conc	0.1416	0.1108	0.1995	0.3378	0.5693	0.5493	0.5967	0.6643	0.1944
FC-Siam-Diff	0.4308	0.3181	0.4826	0.5487	0.7144	0.6564	0.7400	0.7727	0.4800
BIT	0.3885	0.2489	0.3986	0.4092	0.6929	0.6217	0.6979	0.7031	0.3958
ChangeFormer	0.5996	0.4587	0.6289	**0.6612**	0.7990	0.7276	0.8136	**0.8296**	0.6272
DTCDSCN	**0.7799**	0.3665	0.5364	0.4087	**0.8886**	0.6816	0.7674	0.7041	0.5349
EATDer	0.6744	**0.4969**	**0.6639**	**0.6538**	0.8364	**0.7470**	**0.8312**	**0.8262**	**0.6625**
SFEARNet	0.6813	**0.4756**	**0.6446**	0.6117	0.8398	**0.7363**	**0.8216**	0.8052	**0.6431**
DDRL	**0.7107**	0.3154	0.4795	0.3618	**0.8539**	0.6559	0.7389	0.6806	0.4779
WaveDSNet	**0.7775**	**0.5616**	**0.7192**	**0.6691**	**0.8880**	**0.7796**	**0.8590**	**0.8341**	**0.7181**

* Color convention: **best**, **2nd-best** and **3rd-best**.

low-backscatter regions, whereas transformer-based methods tend to oversmooth fine boundaries or miss thin tributaries. Overall, WaveDSNet generally provides more continuous predictions for complex water structures while suppressing many pseudo-change responses, which is consistent with its stronger test-set-level performance.

2) Results on Large-Scale Public Flood Datasets: To further evaluate the proposed method on large-scale public benchmarks, additional experiments are conducted on S1GFloods, VarFloods-P, and VarFloods-G, as reported in Table IV and Fig. 8. All comparison methods are independently retrained on the corresponding training sets and evaluated on the corresponding test sets using the same experimental settings as those adopted for XDU-SWCD. WaveDSNet achieves the best overall IoU, $F1$-score, and recall on S1GFloods, reaching 0.9581, 0.9785, and 0.9831, respectively. On VarFloods-P, it obtains the best overall precision, IoU, and $F1$-score of 0.9808, 0.9611, and 0.9799, while achieving the second-best recall of 0.9790. On VarFloods-G, WaveDSNet achieves the best performance across all four metrics, with precision, IoU, $F1$-score, and recall reaching 0.9858, 0.9704, 0.9848, and 0.9838, respectively. The qualitative comparisons in Fig. 8 are consistent with these quantitative results. On S1GFloods, WaveDSNet produces cleaner change maps

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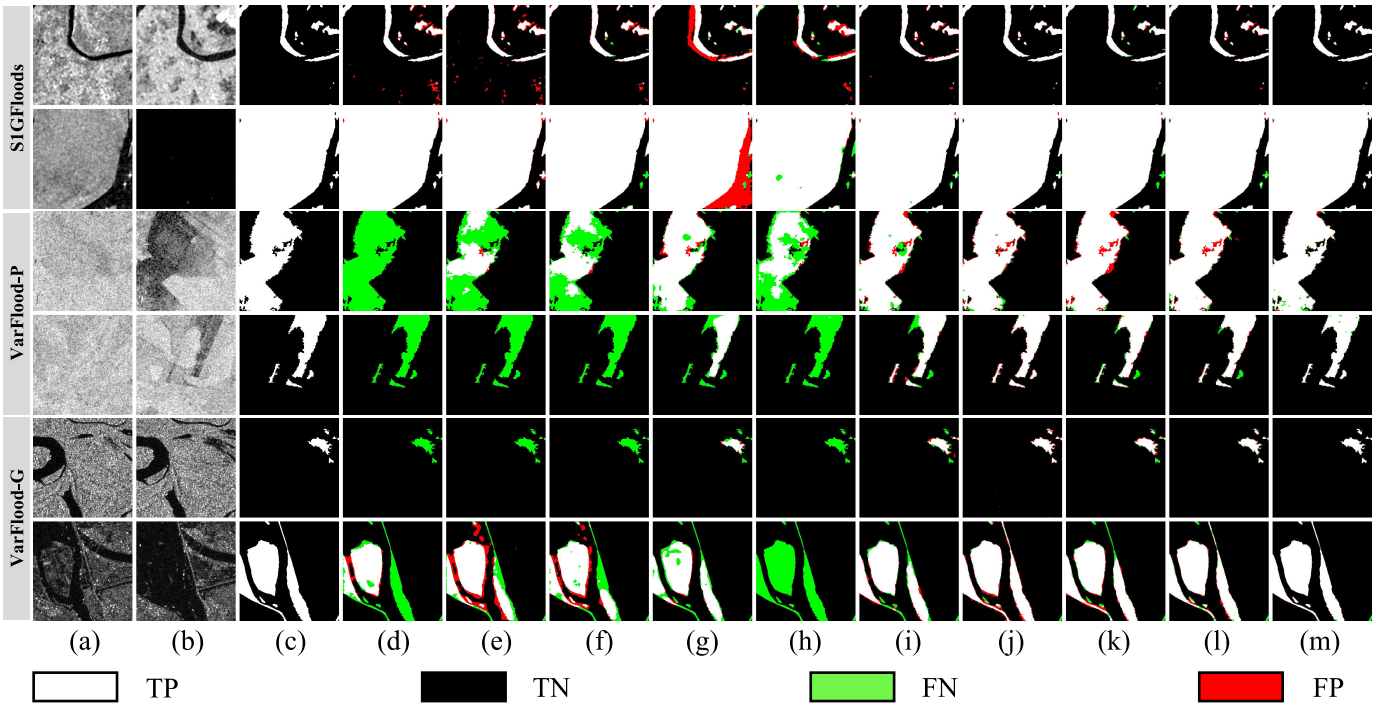


Fig. 8. Visualization comparisons on three large-scale public flood datasets. (a) T1 image. (b) T2 image. (c) GT. (d) FC-EF. (e) FC-Siam-Conc. (f) FC-Siam-Diff. (g) BIT. (h) ChangeFormer. (i) DTCDSCN. (j) EATDer. (k) SFEARNet. (l) DDRL. (m) WaveDSNet.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON S1GFLOODS, VARFLOODS-P, AND VARFLOODS-G

Method	S1GFloods				VarFloods-P				VarFloods-G			
	Precision	IoU	F1-score	Recall	Precision	IoU	F1-score	Recall	Precision	IoU	F1-score	Recall
FC-EF	0.9534	0.9114	0.9533	0.9533	0.4771	0.4771	0.4883	0.5000	0.8838	0.6366	0.7272	0.6658
FC-Siam-Conc	0.9400	0.8943	0.9438	0.9479	0.8560	0.6722	0.7658	0.7145	0.8502	0.6815	0.7758	0.7295
FC-Siam-Diff	0.9471	0.9103	0.9528	0.9592	0.8618	0.6952	0.7888	0.7421	0.8606	0.6820	0.7761	0.7261
BIT	0.8918	0.8219	0.9012	0.9138	**0.9573**	0.7805	0.8626	0.8024	**0.9357**	0.7880	0.8686	0.8208
ChangeFormer	0.8906	0.8055	0.8908	0.8910	0.8178	0.5495	0.6131	0.5742	0.7582	0.4887	0.5126	0.5125
DTCDSCN	0.9487	0.9145	0.9551	0.9625	0.9255	0.8369	0.9042	0.8852	0.9023	0.7996	0.8776	0.8560
EATDer	**0.9767**	**0.9545**	**0.9767**	**0.9767**	0.9481	**0.9345**	**0.9652**	**0.9838**	0.9254	**0.9109**	**0.9516**	**0.9817**
SFEARNet	0.9696	0.9412	0.9696	0.9695	0.9310	0.8801	0.9327	0.9344	0.9242	0.8729	0.9282	0.9324
DDRL	**0.9721**	**0.9487**	**0.9736**	**0.9751**	**0.9568**	**0.9274**	**0.9612**	**0.9656**	**0.9593**	**0.9331**	**0.9644**	**0.9696**
WaveDSNet	**0.9742**	**0.9581**	**0.9785**	**0.9831**	**0.9808**	**0.9611**	**0.9799**	**0.9790**	**0.9858**	**0.9704**	**0.9848**	**0.9838**

* Color convention: **best**, **2nd-best** and **3rd-best**.

with fewer scattered false-positive responses. On VarFloods-P, it recovers more complete changed-water regions, whereas several comparison methods exhibit substantial false-negative areas. On VarFloods-G, WaveDSNet better preserves the continuity of water boundaries while suppressing isolated false activations. These results demonstrate that WaveDSNet maintains strong and stable performance across multiple independently trained public benchmarks and that its effectiveness is not restricted to XDU-SWCD.

3) Zero-Shot Cross-Dataset Transfer Evaluation: To assess cross-dataset transferability, a zero-shot experiment is conducted by training on XDU-SWCD and directly evaluating on 11 conventional public change detection datasets, namely, Bern, Chaohu-1, Ottawa, San Francisco, Sardinia, Sulzberger-1, Sulzberger-2, and Yellow-River-1 ~ 4 without fine-tuning. This setting is highly challenging because the test data differ from XDU-SWCD in sensor characteristics, spatial resolution, geographic regions, and polarization modes, thus introducing substantial domain shifts. Qualitatively, as shown in Fig. 9, most comparison methods produce fragmented predictions or obvious false activations under cross-dataset shifts, whereas WaveDSNet generates more spatially coherent change maps that are closer to the GT. Quantitatively, Table V shows that WaveDSNet achieves the

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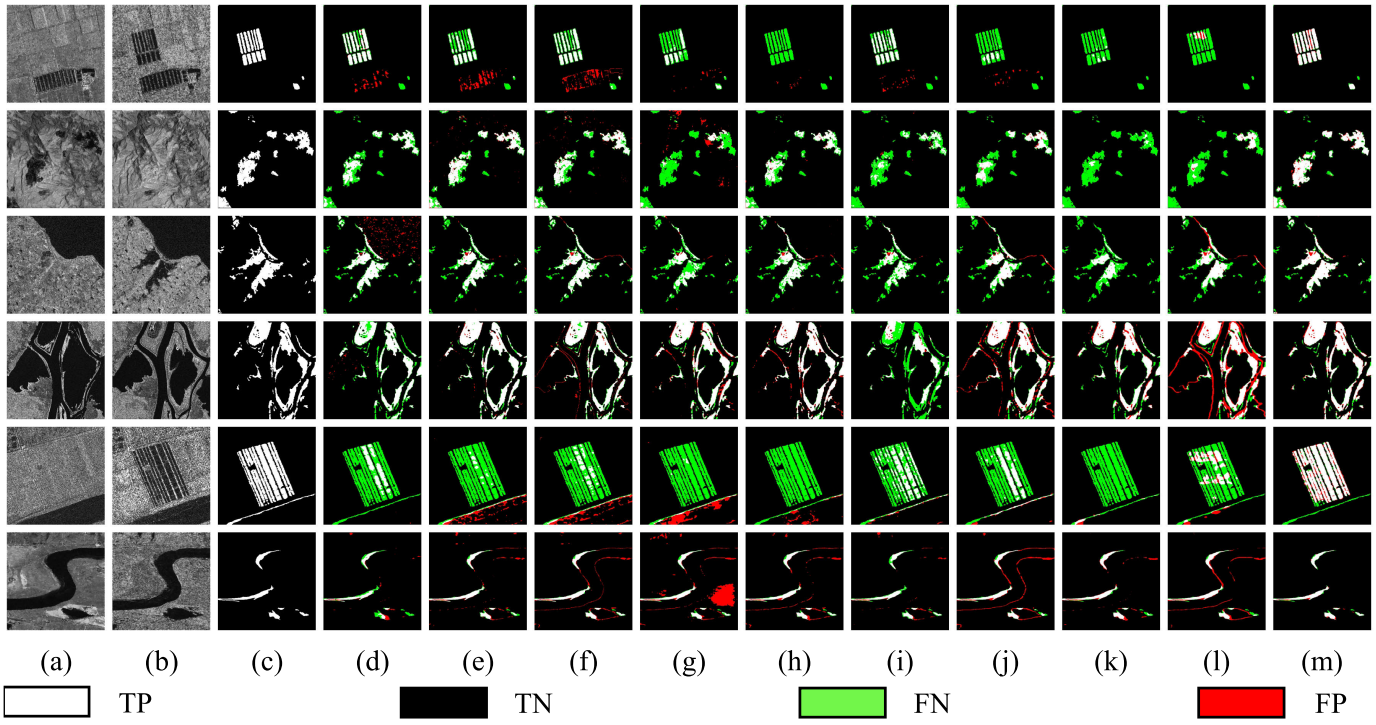


Fig. 9. Visualization comparisons on the public datasets. (a) T1 image. (b) T2 image. (c) GT. (d) FC-EF. (e) FC-Siam-Conc. (f) FC-Siam-Diff. (g) BIT. (h) ChangeFormer. (i) DTCDSCN. (j) EATDer. (k) SFEARNet. (l) DDRL. (m) WaveDSNet.

TABLE V
PERFORMANCE COMPARISON OF CHANGE DETECTION METHODS ON PUBLIC DATASETS

Method	Change Class Metrics				Overall Metrics				
	Precision_1	IoU_1	F1-score_1	Recall_1	Precision	IoU	F1-score	Recall	Kappa
FC-EF	0.9198	0.4319	0.6032	0.4488	**0.9260**	0.6798	0.7829	0.7218	0.5706
FC-Siam-Conc	0.8676	**0.4908**	**0.6584**	**0.5305**	0.9045	**0.7113**	**0.8116**	**0.7599**	**0.6254**
FC-Siam-Diff	0.9036	**0.4852**	**0.6534**	**0.5117**	0.9215	**0.7091**	**0.8094**	**0.7523**	**0.6216**
BIT	0.6819	0.2800	0.4375	0.3221	0.7993	0.5901	0.6925	0.6512	0.3923
ChangeFormer	0.9054	0.4243	0.5958	0.4440	0.9185	0.6753	0.7788	0.7190	0.5624
DTCDSCN	**0.9465**	0.3187	0.4833	0.3245	**0.9324**	0.6175	0.7198	0.6611	0.4508
EATDer	0.8501	0.4172	0.5887	0.4503	0.8911	0.6701	0.7743	0.7199	0.5528
SFEARNet	**0.9350**	0.2594	0.4120	0.2642	0.9234	0.5846	0.6824	0.6309	0.3803
DDRL	0.7564	0.3522	0.5209	0.3972	0.8409	0.6316	0.7372	0.6902	0.4793
WaveDSNet	**0.9702**	**0.7312**	**0.8448**	**0.7480**	**0.9690**	**0.8482**	**0.9135**	**0.8725**	**0.8273**

* Color convention: **best**, **2nd-best** and **3rd-best**.

best performance on all reported metrics, with Precision_1, IoU_1, $F1\text{-score}_1$, and Recall_1 reaching 0.9702, 0.7312, 0.8448, and 0.7480, respectively. Compared with the second-best method, FC-Siam-Conc, WaveDSNet improves IoU_1 and $F1\text{-score}_1$ by 24.04 and 18.64 percentage points, respectively, while Kappa increases from 0.6254 to 0.8273. These results indicate that WaveDSNet generalizes more effectively under severe domain shifts than existing CNN-based, transformer-based, and recent state-of-the-art methods. This advantage suggests that the coordinated design of the WNS-embedded backbone and the SBCR module provides more transferable representations for SAR water change detection across different datasets.

The relatively good zero-shot performance may result from the joint contributions of the diversity of XDU-SWCD and the representation design of WaveDSNet. From the dataset perspective, XDU-SWCD contains diverse temporal change settings, including Dry→Wet, Wet→Wet, and Wet→Dry cases, together with representative hard-negative scenarios. These samples expose the model to different hydrological states, fine-grained water structures, and low-backscatter confounders during training, reducing its dependence on a single water

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TABLE VI
PERFORMANCE COMPARISON OF METHOD COMPONENTS

Method	WNS	SBCR	Change-Class Metrics				Overall Metrics				
			Precision_1	IoU_1	F1-score_1	Recall_1	Precision	IoU	F1-score	Recall	Kappa
Baseline			0.6871	0.3979	0.5693	0.4860	0.8429	0.6981	0.7842	0.7427	0.5684
Baseline	✓		0.6230	0.4835	0.6518	**0.6834**	0.8111	0.7408	0.8254	**0.8412**	0.6509
Baseline		✓	0.6887	0.4856	0.6537	0.6221	0.8439	0.7420	0.8264	0.8107	0.6529
Baseline	✓	✓	**0.7775**	**0.5616**	**0.7192**	0.6691	**0.8880**	**0.7796**	**0.8590**	0.8341	**0.7181**

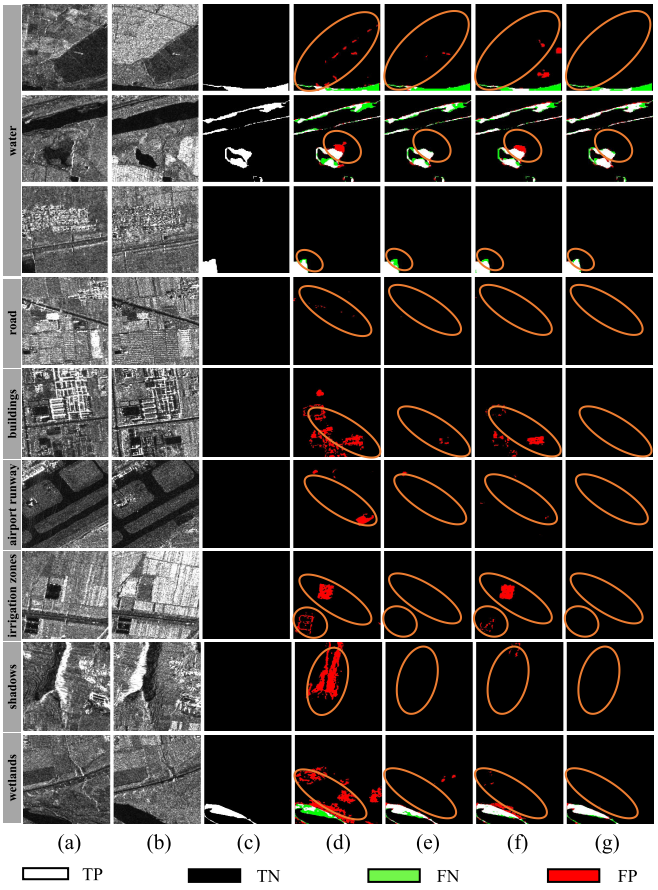


Fig. 10. Qualitative ablation visualization on nine representative examples covering seven scenario types of XDU-SWCD. (a) T1 image. (b) T2 image. (c) GT. (d) Baseline. (e) Baseline+WNS. (f) Baseline+SBCR. (g) WaveDSNet.

appearance or background distribution. From the model perspective, the WNS-embedded backbone performs localized multiscale refinement in wavelet subbands, which may reduce the sensitivity of the learned features to dataset-specific radiometric fluctuations while preserving structural responses. Meanwhile, SBCR adaptively coordinates relative temporal discrepancy cues and semantic-structural information instead of relying exclusively on raw temporal differences. This complementary modeling may improve robustness when sensor characteristics and imaging conditions change. Nevertheless, neither WNS nor SBCR explicitly eliminates domain shift, and the observed zero-shot performance should be interpreted as improved cross-dataset robustness rather than complete domain invariance.

C. Ablation Study

1) Hierarchical WNS Ablation Study: To evaluate the impact of hierarchical WNS integration, SBCR is kept enabled while WNS is progressively embedded from Layer 1 to Layer 4, as shown in Table VII. Compared with the SBCR-enabled baseline, adding WNS at Layer 1 improves $F1\text{-score}_1$ and IoU_1 from 0.6537 and 0.4856 to 0.6670 and 0.5004, respectively. Extending WNS to Layer 2 increases Recall_1 to 0.6733, while deeper integration consistently improves $F1\text{-score}_1$ and IoU_1. The full Layer 1–Layer 4 configuration achieves the best overall performance, with $F1\text{-score}_1$, IoU_1, and Kappa reaching 0.7192, 0.5616, and 0.7181, respectively, demonstrating the cumulative benefit of hierarchical WNS integration.

2) Module-Level Ablation: Module-level ablation experiments are first conducted on the full test set, as reported in Table VI. Compared with the baseline, WNS improves $F1\text{-score}_1$, IoU_1, and Recall_1 from 0.5693, 0.3979, and 0.4860 to 0.6518, 0.4835, and 0.6834, respectively, indicating its effectiveness in enhancing weak changed-water responses under speckle interference. SBCR also improves $F1\text{-score}_1$ and IoU_1 to 0.6537 and 0.4856, while achieving higher Precision_1 than WNS, showing its ability to suppress pseudo-changes through semantic-difference interaction and adaptive gating. By combining WNS and SBCR, WaveDSNet achieves the best $F1\text{-score}_1$, IoU_1, and Kappa of 0.7192, 0.5616, and 0.7181, respectively. The qualitative results in the first three rows of Fig. 10 are consistent with the quantitative results and show that WNS reduces fragmented and speckle-induced false activations, and SBCR improves the spatial coherence and localization of changed-water regions.

To further evaluate robustness against hard-negative samples, a scenario-level subset is constructed from the test set, containing 20 image pairs for each of the six scenarios. Since these scenarios correspond to confusing unchanged regions rather than independently annotated semantic classes, category-specific IoU or Kappa cannot be directly calculated. Moreover, binary IoU and Kappa computed over the complete image pairs would jointly reflect the detection of true changed-water regions and unchanged background regions, and therefore cannot adequately characterize the false-alarm tendency associated with different hard-negative scenarios. Accordingly, the false-positive rate (FPR) is adopted as a

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TABLE VII
IMPACT OF HIERARCHICAL WNS INTEGRATION ON XDU-SWCD WITH SBCR ENABLED

Layer1	Layer2	Layer3	Layer4	Change Class Metrics				Overall Metrics				
				Precision$_1$	IoU$_1$	F1-score$_1$	Recall$_1$	Precision	IoU	F1-score	Recall	Kappa
-	-	-	-	0.6887	0.4856	0.6537	0.6221	0.8439	0.7420	0.8264	0.8107	0.6529
✓	-	-	-	0.7595	0.5004	0.6670	0.5947	0.8788	0.7489	0.8329	0.7969	0.6657
✓	✓	-	-	0.6660	0.5033	0.6696	**0.6733**	0.8323	0.7502	0.8340	**0.8359**	0.6681
✓	✓	✓	-	0.7029	0.5154	0.6802	0.6590	0.8507	0.7563	0.8394	0.8289	0.6789
✓	✓	✓	✓	**0.7775**	**0.5616**	**0.7192**	0.6691	**0.8880**	**0.7796**	**0.8590**	0.8341	**0.7181**

TABLE VIII
SCENARIO-WISE FPRS ON THE HARD-NEGATIVE SUBSETS OF XDU-SWCD

Scenario	FPR (%) ↓			
	Baseline	Baseline + WNS	Baseline + SBCR	WaveDSNet
Roads	3.03	0.06	0.38	**0.01**
Buildings	6.52	0.34	1.01	**0.16**
Airport Runways	1.53	0.08	0.20	**0.02**
Irrigation Zones	3.90	0.30	1.94	**0.10**
Shadows	8.27	0.94	1.42	**0.28**
Wetlands	10.29	0.29	0.78	**0.02**

FPR denotes the percentage of unchanged pixels incorrectly classified as changed pixels. Lower values indicate fewer false alarms. WaveDSNet denotes the complete model equipped with both WNS and SBCR.

scenario-conditioned measure to quantify the proportion of unchanged pixels incorrectly predicted as changed within each corresponding subset

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \times 100\% \tag{23}$$

where FP denotes the number of GT unchanged pixels incorrectly classified as changed, TN denotes the number of GT unchanged pixels correctly classified as unchanged, and FP + TN is the total number of GT unchanged pixels in the corresponding scenario-specific subset. Table VIII shows that the baseline produces substantial false positives, particularly for wetlands, shadows, and buildings, with FPRs of 10.29%, 8.27%, and 6.52%, respectively. Both WNS and SBCR reduce false responses across all six scenarios, while the complete WaveDSNet achieves the lowest FPR in every scenario. The corresponding qualitative results in the last six rows of Fig. 10 further confirm that WNS and SBCR complement each other in suppressing diverse hard-negative responses.

D. Sensitivity Analysis of the Boundary Loss Weight

To analyze the influence of the boundary loss weight λ, a sensitivity analysis is conducted on the validation set of XDU-SWCD, as reported in Table IX. As λ increases from 0.1 to 0.7, precision, IoU, and $F1$-score improve consistently, indicating that appropriate boundary-aware auxiliary supervision benefits the main water change detection task. Although $\lambda = 0.1$ yields the highest recall, its lower precision, IoU, and $F1$-score indicate a less balanced precision–recall tradeoff. When $\lambda = 0.7$, the model achieves the best precision, IoU, and $F1$-score, reaching 0.8517, 0.7566, and 0.8419, respectively. Further increasing λ to 0.9 reduces the performance, suggesting that excessive boundary supervision may interfere with the optimization of the main change detection task. Therefore, $\lambda = 0.7$ is adopted in the final experiments.

TABLE IX
SENSITIVITY ANALYSIS OF THE BOUNDARY LOSS WEIGHT

λ	Precision	IoU	F1-score	Recall
0.1	0.8052	0.7436	0.8310	**0.8620**
0.3	0.8075	0.7446	0.8318	0.8607
0.5	0.8347	0.7534	0.8369	0.8233
0.7	**0.8517**	**0.7566**	**0.8419**	0.8494
0.9	0.8419	0.7485	0.8348	0.8281

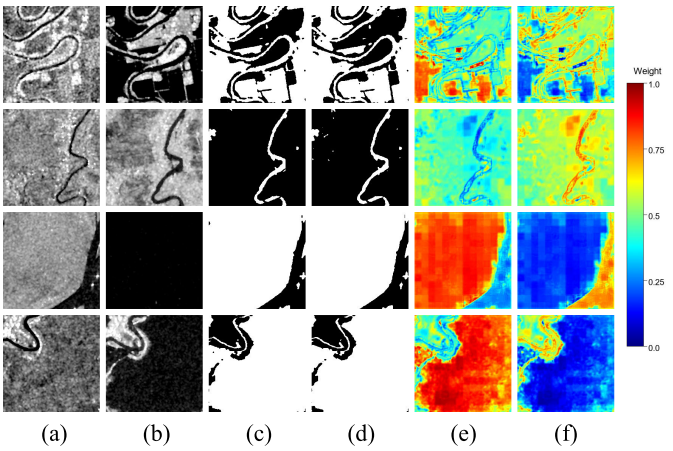


Fig. 11. Visualizations of adaptive gating weights in the SBCR module under different water change patterns. (a) T1 image. (b) T2 image. (c) GT. (d) Prediction result. (e) Heatmap of the adaptive weight W_1 for the refined discrepancy stream F'_D. (f) Heatmap of the adaptive weight W_2 for the refined semantic stream F'_S. Warmer colors indicate larger weights.

E. Visualization of Adaptive Gating Weights

To further interpret the adaptive fusion mechanism of SBCR, the heatmaps of W_1 and W_2 are visualized in Fig. 11. Here, W_1 and W_2 denote the spatially adaptive weights assigned to the refined discrepancy stream F'_D and the refined

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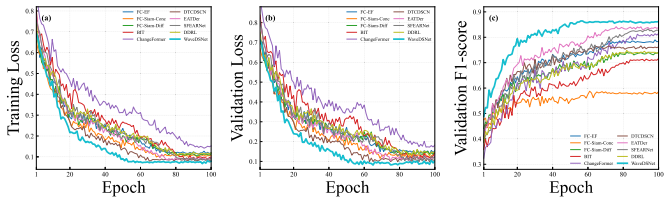


Fig. 12. (a) Training loss, (b) validation loss, and (c) validation $F1$-score curves of different methods over 100 training epochs.

semantic stream F'_S, respectively. Fig. 11 shows that the two weight maps exhibit complementary response patterns. For broad water changes with clear temporal discrepancies, W_1 shows stronger responses over the changed regions, indicating a greater contribution from the refined discrepancy stream. For narrow river-like changes, W_2 shows stronger responses along thin water structures, suggesting that the refined semantic stream provides complementary structural cues when local discrepancy responses are less stable. These patterns suggest that the SBCR gating mechanism adaptively balances the two refined streams according to the scale and spatial structure of water changes.

F. Convergence Analysis

Fig. 12 shows the training loss, validation loss, and validation $F1$-score curves of all compared methods over 100 epochs. Most methods reach relatively stable training and validation losses after approximately 60–80 epochs, with only moderate validation fluctuations and no obvious late-stage divergence. Their validation $F1$-scores also approach stable plateaus, while WaveDSNet reaches a stable high-performance region after approximately 50–60 epochs and maintains the highest validation $F1$-score thereafter. These results indicate that all methods converge within 100 epochs, with WaveDSNet showing stable and relatively rapid convergence.

VI. Discussion and Limitations

A. Tradeoff Between False Positives and False Negatives

WaveDSNet achieves stronger quantitative performance on the complete test set and one important aspect of this advantage lies in its effective suppression of false-positive responses. Nevertheless, local classification errors can still occur in some scenes, particularly in the fourth and final rows of Fig. 7, where SFEARNet preserves relatively more complete changed-water regions, whereas WaveDSNet produces more conservative predictions with false negatives. A possible reason is that WaveDSNet places strong emphasis on suppressing pseudo-change responses caused by speckle interference and confusing low-backscatter targets. When true changed regions also exhibit locally weak or unstable discrepancy responses, the model may assign relatively low confidence to the change class and consequently classify some true changes as unchanged. This behavior indicates a local tradeoff between false-positive suppression and the preservation of uncertain true-change responses. Future work will investigate more adaptive decision and confidence-calibration strategies to better balance false-positive suppression and missed-detection control while preserving robustness against pseudo-change responses.

TABLE X
Efficiency Comparison of Different Methods on XDU-SWCD

Method	Params (M)	FLOPs (G)	Latency (ms)	FPS
FC-EF	1.35	3.12	2.24	446.50
FC-Siam-Conc	1.55	4.88	3.20	312.05
FC-Siam-diff	1.35	4.27	3.15	317.91
BIT	11.85	26.00	3.87	258.73
DDRL	34.59	42.55	56.47	17.71
ChangeFormer	38.20	9.94	19.21	52.05
DTCDSCN	41.07	11.50	12.60	79.37
EATDer	7.12	20.79	20.85	47.96
SFEARNet	5.56	4.65	43.27	23.11
WaveDSNet	59.24	47.04	77.47	12.91

B. Computational Efficiency and Deployment Limitations

From the perspective of computational efficiency, WaveDSNet still incurs a relatively high deployment cost. Table X shows that WaveDSNet contains 59.24M parameters and requires 47.04-G FLOPs, with an inference speed of 12.91 FPS, resulting in higher computational cost than most comparison methods. This overhead is associated with the hierarchical CSWin transformer backbone, multistage WNS refinement, and bidirectional feature interaction in SBCR. Specifically, WNS repeatedly performs wavelet decomposition, subband-wise dynamic convolution, and feature reconstruction across multiple feature levels. Meanwhile, SBCR introduces bidirectional cross-stream interaction and adaptive fusion, further increasing computational and inference costs while improving representation capability in complex SAR scenes. This computational overhead may limit the deployment efficiency of WaveDSNet in real-time disaster-response and resource-constrained scenarios. Future work will focus on lightweight backbone architectures, selective WNS integration, and more efficient cross-stream interaction to reduce redundant computation while preserving the robustness of WaveDSNet against pseudo-changes and complex water structures.

VII. Conclusion

This article presents WaveDSNet for water change detection in SAR imagery and introduces XDU-SWCD as a focused high-resolution benchmark for evaluation. WaveDSNet consists of a WNS-embedded backbone and an SBCR module. The former performs wavelet-subband-aware denoising to suppress speckle while preserving structural cues, whereas the latter coordinates semantic features and difference features through bidirectional interaction and adaptive gating, and further imposes boundary-aware supervision for fine-grained changed-water boundary localization. Experiments on XDU-SWCD show that the proposed method achieves stronger overall performance than representative CNN-based, transformer-based, and recent state-of-the-art change detection

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models. Zero-shot cross-dataset evaluation on public benchmarks further indicates the promising transferability of the framework. Overall, the results suggest that effective water change detection in SAR imagery requires not only speckle-robust representation learning, but also adaptive coordination between semantic detail and discrepancy evidence, together with boundary-aware regularization. Future work will focus on improving the balance between false-positive suppression and false-negative reduction through more adaptive decision strategies, while reducing computational overhead through lightweight backbone design, selective WNS integration, and more efficient cross-stream interaction.

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